Spectro-textural Integration in Mangrove Delineation: A Case Analysis of Aboitiz Cleanergy Park, Davao City, Philippines

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***Abstract*—Mangroves, located between land and sea, are highly biodiverse ecosystems that provide services, enhancing coastal resilience against climate change. Despite their importance, mangroves face significant degradation due to human activities, highlighting the need for detailed studies on their spatial extent. This study evaluates remote sensing (RS) and geographical information systems (GIS) techniques for delineating mangroves in Aboitiz Cleanergy Park, Punta Dumalag, Davao City, Philippines. The Park is selected for its ecological significance as a protected area with diverse coastal habitats, including mangroves. Sentinel-2A images are processed using GIS and the Sentinel Application Platform (SNAP). Mangrove areas are classified using Google Earth Pro imagery and ground truthing. Vegetation indices, principal components, and Grey Level Co-Occurrence Matrix (GLCM) textures are analyzed. A Random Forest algorithm achieves robust classification, yielding a kappa value of 0.896. Results reveal a 42.16% increase in mangrove coverage from 2019 to 2024. The study demonstrates the effectiveness of RS in accurately mapping mangroves and supporting sustainable management and conservation efforts.**

Keywords—Geographic Information System, Grey Level Co-occurrence Matrix, Mangrove Delineation, Principal Component Analysis, Vegetation Index

I. Introduction

Mangroves are an ecosystem between the land and the sea that can tolerate tidal fluctuations and are found at low latitudes in tropical or subtropical regions. It is widely recognized as one of the primary traces of defense against environmental risks [1]. Regarding ecology, mangroves protect the coast from solar UV-B radiation, "greenhouse" gasses, storms, floods, rising sea levels, wave action, and coastal soil erosion. It also serves as a food web-supporting source, sinks, and traps for nutrients. They are best at mitigating the effects of climate change and storing carbon. Numerous food fish and wildlife creatures use them as feeding, breeding, and nidification sites. Mangroves also guard various aquatic ecosystems, including seaweed beds, coral reefs, islands, and seagrass meadows [2]. Despite their significance, mangrove ecosystems are increasingly threatened by human activities.[3] It is disappearing globally at an alarming rate; in the last 20 years, at least 23% of mangrove forests have vanished [4].

As an archipelago country with more than 7,000 islands, the Philippines possesses one of the world's longest coastlines, up to 36,289 km. It is situated at 13°00°N, 122°00°E, near the tropical band where mangroves are abundant. Due to its geographic location, mangrove diversity is relatively high [5]. However, because of human activities, mangrove areas in the Philippines have been disappearing over time [6]. Only 15 of 81 provinces in the Philippines have mangrove forest areas designated and recognized as protected areas [6]. Awareness and understanding of the mangrove ecosystem pattern and their services is essential for conservation and management; this aspect is still understudied and less explored, particularly in the Philippines [7].

To quantify mangrove forests' services, mapping their natural extent is essential. Mangroves face danger from deforestation and climate change, so extensive mangrove mapping is required to analyze these threats. Conventional methods for detecting mangroves have limitations in accuracy, efficiency, and automation, posing challenges for comprehensive environmental assessment [8]. Most studies use remote sensing technologies to measure and identify the extent of mangrove forests. Many methodologies have been developed to delineate mangrove from non-mangrove vegetation [9]. Implementing and applying multi-spectral remote sensing images is crucial to better understanding the earth's environment. Remote sensing is the scientific acquisition of information and extraction features in spectral, spatial, and temporal terms regarding a specific area, such as land cover classification and tracking vegetation changes [10]. It provides accurate information in the past and present regarding the status of mangroves in a specific area with high temporal, spatial, and spectral resolution subject to qualitative and quantitative assessments [11]. However, the quality of the images acquired

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should be assessed to provide a more accurate and comprehensive description of the mangrove's status. Many remote sensing studies have been conducted to assess land use and forest area coverage changes, yet distinguishing mangroves from non-mangrove vegetation is still incomprehensive and lacks foundation. Analysis and management of mangrove degradation based on their structure is made possible by integrating and incorporating remote sensing techniques and spatial analysis [12].

By accurately mapping out the extent and distinction of mangroves, mangrove monitoring would be less complex, allowing more comprehensive assessments regarding the dynamics and health of mangroves. This study utilizes advanced remote sensing techniques and spatial metrics to distinguish mangroves from non-mangrove vegetation. Therefore, this study's main objectives are to (i) distinguish mangroves from non-mangrove vegetation, (ii) enhance the visibility of satellite images, making specific details more obvious and (iii) assess the accuracy of the classified maps by comparing them to high-resolution satellite images.

II. Materials and Methods

A. Description of the Study Area

This study employs a quantitative research approach that follows a descriptive research design. Systematic empirical data-gathering methods and statistical analysis are observed to answer research questions. Descriptive research design describes the classification of mangroves.

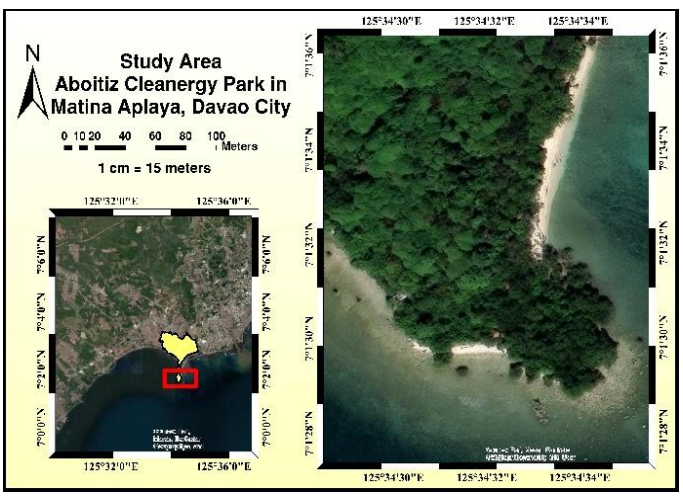

Fig. 1. Study Area (Matina Aplaya, Davao City, Philippines).

This study focuses on Matina Aplaya, Davao City, Davao del Sur, with coordinates approximately 7°3' N latitude and 125°34' E longitude. According to the 2020 Census of Population, as of June 30, 2024, Matina Aplaya is a barangay in Davao City and has a total population of 32,396 comprising 1.82% of the total population of Davao City (Philippine Statistics Authority, 2020). Specifically at Punta Dumalag, Matina Aplaya, with an eight-hectare Biodiversity Park, Aboitiz Cleanergy Park is a nesting ground for various aviaries, managed by AboitizPower's Subsidiary, Davao Light, and Aboitiz Foundation Inc. Established in 2014, this conservation area encompasses a turtle habitat, coastal forests, mangroves, and a coral reef ecosystem. Surrounded by a 37-hectare Marine Protected Area (MPA) dedicated to preserving marine biodiversity [13].

B. Data Collection and Processing

The study employs indirect methods of data collection to gather spatial data. Satellite images are obtained explicitly from the Copernicus Data Space web portal. Satellite time-series imagery from sentinel-2 is used to map and track the expansion or contraction of mangrove cover. Specifically, B2, B3, B4, B5, B6, B7, B8, B8a, B11, and B12 are downloaded from the website to generate vegetation indices and Gray-Level Co-Occurrence Matrix texture. These are used to distinguish mangrove vegetation from non-mangrove vegetation.

The adoption of GIS principles for mangrove classification adheres to the methodology proposed by Sunkur et al. (2024) [14]. Bands 2, 3, 4, and 8 feature a 10 m × 10 m spatial resolution, while bands 5, 6, 7, 8A, 11, and 12 contain a 20 m × 20 m spatial resolution. Using QGIS software, the bands undergo resampling to increase the spatial resolution from 20 m to 10 m through the nearest neighbor resampling technique. The products then undergo reprojection to WGS 1984 UTM Zone 51N, aligning with the boundaries of the study area. Finally, the resampled and reprojected bands undergo clipping based on the extent of the study area.

To extract spectral features from the satellite images, four vegetation indices are calculated such as:

$$NDVI = \frac{B8 - B4}{B8 + B4} \tag{1}$$

$$EVI\,2 = 2.5 * \frac{B8 - B4}{B8 + (2.4 * B4) + 1} \tag{2}$$

$$GNDVI = \frac{B8 - B3}{B8 + B3} \tag{3}$$

$$ReNDVI = \frac{B6 - B5}{B6 + B5} \tag{4}$$

PCA (Principal Component Analysis) is implemented to remove redundant spectral data from multiband datasets. Next, correlation, entropy, and homogeneity mean are extracted using the Grey Level Co-Occurence Matrix (GLCM) statistical technique. GLCM evaluates the properties of pixel pairing with respect to a specific distance and position, which effectively represents the spatial dependence between the values of spectral features, resulting in a comprehensive assessment of the spatial distribution and density of vegetation [15]. To derive the four textural variables on SeNtinel Application Platform (SNAP), a 5x5 window is applied and cropped to the extent of the study area, setting the probabilistic quantizer and quantization levels to 32 and All angles, respectively. The resulting images stack to generate a 26-feature composite image composed of 10 Sentinel 2A bands, 4 vegetation indices, and 12 textural features. Random Forest Classification is implemented to enhance the classification of mangrove forests further and generate accurate maps showing the classified mangrove cover. Finally, a Stratified Random Strategy is employed for the accuracy

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assessment of the classification. This statistical technique generates points randomly distributed across each class, and the number of points each class contains is relative to the class's area. Confusion matrix is used to calculate incorrectly classified images in each class in which kappa analysis is conducted to assess the spectral signatures of the classified data.

C. Data Sources

Researchers consult the City Planning and Development Office (CPDO) of Davao City, Philippines for the acquisition of Davao City barangay boundaries. In this study, the researchers approached a government agency, specifically the City Planning and Development Office (CPDO- Davao City) to acquire specific spatial data. The satellite time-series imagery from sentinel-2 from the year 2024 are used to map and track the expansion or contraction of mangrove cover. Acquired satellite images have a cloud cover of 6.25%. Acquired satellite image data are accessed from Copernicus Browser (https://browser.dataspace.copernicus.eu). Sentinel-2 is a European mission that utilizes wide-swath, high-resolution, multi-spectral imaging. The mission supports operational applications primarily for land services, including the monitoring of vegetation, soil and water cover, as well as the observation of inland waterways and coastal areas. The twin satellites fly in the same orbit but are phased at 180°, and the full mission specification is designed to provide a high revisit frequency of 5 days at the Equator [16].

TABLE I. DATA AND SOURCES

Data	***Sources***
Davao City barangay boundary	City Planning and Development Office (Davao City)
Bands: B2, B3, B4, B5, B6, B7, B8, B8A, B11, AND B12	Sentinel 2 (Copernicus Browser)

D. Methodological Framework

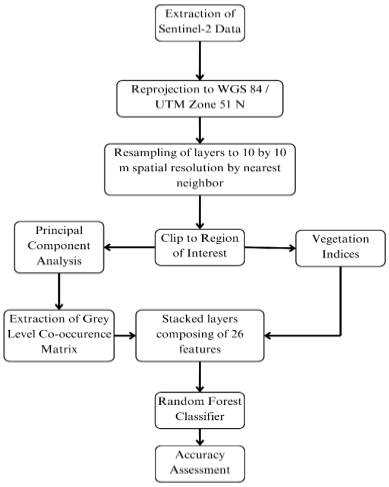

Fig. 2. Methodological Framework of the study.

III. Results and Discussions

This section analyzes and presents the results of the methodology, focusing on the Application of PCA, land cover classification is derived using Random Forest classification, and accuracy assessment including confusion matrix and kappa value indicating performance classification.

A. Field Observations

Ground data are acquired using the Global Positioning System (GPS) to validate the classified map. At Aboitiz Cleanergy Park, many organizations have initiated mangrove restoration projects, which result in dense mangrove vegetation in the southern part of the park. The landward fringe consists mainly of old and mature mangroves. The existence of newly planted seedlings and growing mangroves are seen along the sandy part of the park. Standing dead corals are spotted near the shoreline beside the dense mangrove vegetation. A minimum of 30 points in each of 5 different classes - mangrove, non-mangrove tree, sand, shallow water, and deep water - are observed to generate a training sample and produce a mangrove classification map. To validate the data, high-resolution satellite images from Google Earth Pro are used to calculate accuracy parameters.

Fig. 3. Field Observation.

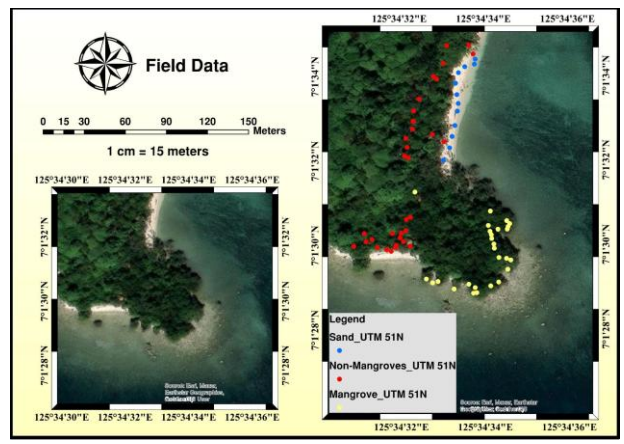

Fig. 4. Field Data.

B. Field Analysis

The Principal Component Analysis (PCA) performed on Sentinel images reduced the original 10 input layers to 3 principal components (PCs), capturing 100% of the variance. As shown in Table 2, PC1 explains most of the variance (91.14%), followed by PC2 (8.05%) and PC3 (0.81%). This feature extraction is critical in enhancing the visibility of satellite images, making specific details more obvious. The eigenvalues reflect the percentage of variance captured by each PC, with higher eigenvalues, indicating more data explained by that component. PC1 captured the largest amount of variance, followed by PC2 and PC3, making PC1 the most important for both sites. Overall, the first two PCs account for 99.19% of the

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variance, indicating that the dataset can be effectively represented with minimal information loss.

TABLE II. PRINCIPAL COMPONENT CHARACTERISTICS

Principal Components	*Eigenvalue*	*% Variance*	*Cumulative % Variance*
PC1	3,205,728.51	91.14	91.14
PC2	282,961.44	8.05	99.19
PC3	28,496.43	0.81	100.00

C. Image Classification

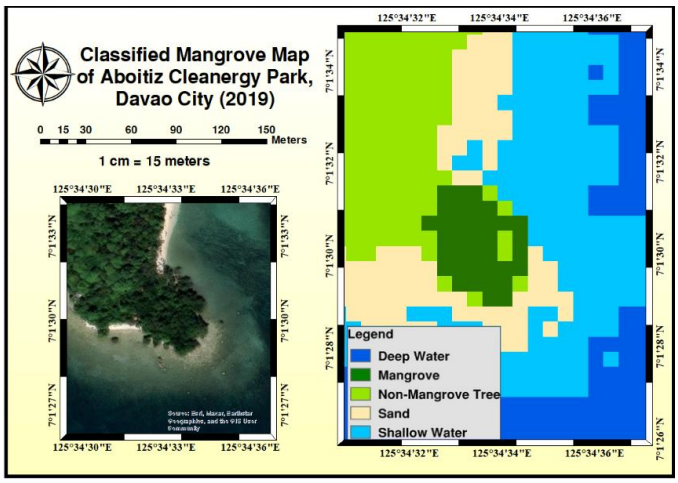

Fig. 5. Classified mangrove cover at Aboitiz Cleanergy Park (2019).

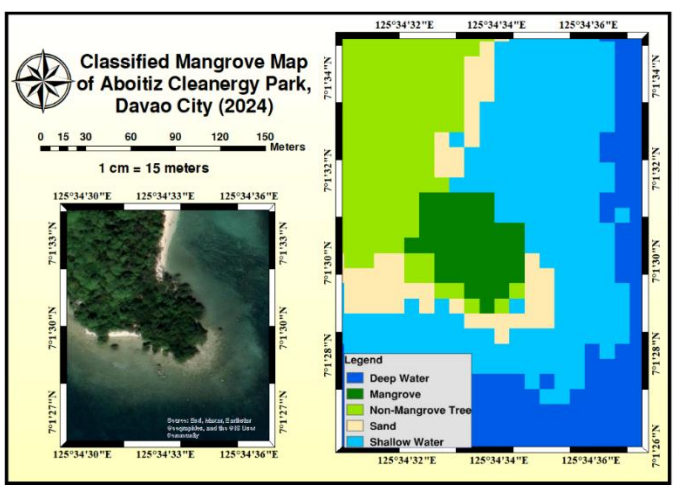

Fig. 6. Classified mangrove cover at Aboitiz Cleanergy Park (2024).

Figure 5 illustrates the classified mangrove cover at Aboitiz Cleanergy Park for 2024. Land cover classes are (i) mangroves with a distinct spectral signature, (ii) non-mangrove tree referring to vegetation other than the mangrove itself, (iii) sand representing sandy areas as well as the standing dead coral, (iv) shallow water representing shallow water areas, and (v) deep water showing deep sea. Results derived from this research show that the mangrove cover in Aboitiz Cleanergy Park increased from 0.37 hectares in 2019 to 0.526 hectares in 2024, representing a 42.16% expansion. Although relatively small in area, this growth is ecologically significant, reflecting the effectiveness of ongoing conservation and rehabilitation efforts. Given the park's strategic location and ecological value, these findings highlight the need for continued protection and informed management.

D. Accuracy Assessment

The confusion matrix in Table 3 shows the classification results across five classes: Mangrove (M), Non-Mangrove Tree (NTM), Sand (Sn), Shallow Water (SW), and Deep Water (DW). The system performs well overall, as shown in Table 2, with most predictions along the diagonal, indicating correct classifications: M (8), NMT (19), Sn (18), SW (19), and DW (26). However, some misclassifications are evident, such as 2 M instances incorrectly predicted as NMT and 1 NMT instance predicted as M. Sn has 1 instance misclassified as SW, while SW and DW show minimal confusion. The user accuracy (U.Accuracy) for each class is high, ranging from 0.800 for M to 0.950 for NMT. The kappa value of 0.896 indicates a robust classification performance.

TABLE III. CONFUSION MATRIX

Class	*M*	*NMT*	*Sn*	*SW*	*DW*	*Total*	*U. Accuracy*	*Kappa*
M	8	2	0	0	0	10	0.800	0
NMT	1	19	0	0	0	20	0.950	0
Sn	0	0	18	1	0	19	0.947	0
SW	0	1	1	19	0	21	0.905	0
DW	0	0	0	3	26	29	0.897	0
Total	9	22	19	23	26	99	0	0
P. Accuracy	0.889	0.864	0.947	0.826	1.00	0	0.908	0
-	0	0	0	0	0	0	0	0.896

The integration of Sentinel-2 imagery and advanced remote sensing techniques in this study successfully classified the mangrove vegetation in Aboitiz Cleanergy Park. Many studies utilize Sentinel-2 imagery in mangrove mapping due to its spectral and spatial resolution [17]. This study managed to address the difficulty and challenge of distinguishing mangrove trees from non-mangrove vegetation by employing vegetation indices and texture metrics through Random Forest classification. Vegetation indices like NDVI, EVI2, GNDVI, and ReNDVI help accurately capture the spectral characteristics of mangrove vegetation especially when combined with GLCM-based texture variables which added spatial context to spectral

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data. [17] emphasizes the importance of extracting GLCM texture features to improve classification accuracy. In this study, these textures provide more information about objects that have similar spectral characteristics, and with the inclusion of GLCM-base texture, the classification accuracy has greatly improved. Correlation measures the linear relationship between pixel values, while entropy evaluates the randomness or complexity of the texture [18]. Homogeneity indicates how similar pixel pairs are, and the mean calculates the average pixel value based on how often it appears with neighboring pixels [19]. According to [20], mangrove mapping can be performed using various techniques, such as Google Earth Engine (GEE). These cloud computing platforms simplify many processes, including the direct retrieval of satellite imagery stored in the cloud, which aids in data access and sharing. However, these platforms are based on coding and require programming skills for conducting remote sensing analyses. Individuals unfamiliar with coding may encounter difficulty when trying to use these platforms. To overcome this matter, this study employed GIS software for remote sensing analysis. Although GIS is often overshadowed by more specialized software in remote sensing applications, it is still a powerful tool for carrying out such studies. It features a user-friendly interface, enabling researchers and practitioners to easily access and analyze remote sensing data without the need for programming expertise, offers an extensive array of tools including image pre-processing, classification, and spatial analysis, and allows for the integration of remote sensing data with other geospatial information [14].

The result of high user accuracy for mangroves (80%) and other classes, such as non-mangrove trees (95%), confirms the reliability of the methodology of this study for regional-scale mangrove classification. The use of PCA to minimize data dimensionality also aided in improving computational efficiency while preserving the integrity of the information. This method streamlined the dataset while retaining essential spatial and spectral information, accounting for 99.19% of the variance in the top two principal components. Nonetheless, some misclassifications are noted in the confusion matrix. These inaccuracies might be attributed to spectral similarities between mangroves and other types of vegetation, along with environmental factors influencing reflectance. It is believed that the modeling process introduced errors due to overlapping characteristics between mangroves, non-mangrove trees, and grassland classes resulting from their similar spectral profiles.

This study also emphasizes the importance of integrating remote sensing with ground validation. Ground truthing conducted at the Aboitiz Cleanergy Park verified the presence of mangroves, demonstrating the need for such data to validate the satellite-derived classifications. Although the methodology of this study demonstrated a reliable classification technique, more research into other machine learning techniques or the use of Google Earth Engine for cloud-based analysis could also further improve the efficacy of mangrove vegetation classification. Notably, the pixelated appearance of the classified maps reflects the small extent of the study area. Expanding the scope to include larger areas and conducting temporal analysis could also improve the study.

IV. Conclusion

This study effectively integrates Sentinel-2 imagery, vegetation indices, and Gray Level Co-Occurrence Matrix (GLCM) texture metrics with Random Forest classification to distinguish mangrove from non-mangrove vegetation. Principal Component Analysis (PCA) is employed, adding textural features resulting in a significant improvement in the computational accuracy of the classification. Additionally, integrating vegetation indices, such as NDVI, EVI2, GNDVI, and ReNDVI, effectively captured the complex characteristics of mangrove vegetation cover. Ground truthing further refined the classified maps, adding essential features by integrating satellite data with field data for accurate mapping and classification. In 2019, the mangrove cover was measured at 3,700 square meters (0.37 hectares). By 2024, it increased by 42.16%, expanding to approximately 5,264.86 square meters or 0.526 hectares. The study's accuracy assessment, with a kappa value of 0.896, emphasized the robustness of the classification methodology.

V. Recommendation

The use of high spectral, spatial and temporal resolution for satellite images are highly encouraged to acquire more accurate results and by combining these advanced techniques for imaging, enhanced precision and accuracy will be achieved in mapping and monitoring mangroves.

Exploring the feasibility of Google Earth Engine (GEE) Platform and other machine learning or classification tools such as Random Forest (RF) and Support Vector Machine (SVM) in monitoring and mapping mangroves. Integrating data from Sentinel-2 could be explored to improve the accuracy of forest maps. Common algorithms used in studies are SVM and RF which are used to enhance classification and mapping of the mangrove ecosystem. Incorporating more vegetation classes can differentiate mangroves from other classification types easily and expanding the number of classification categories results in more significant distinction.

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